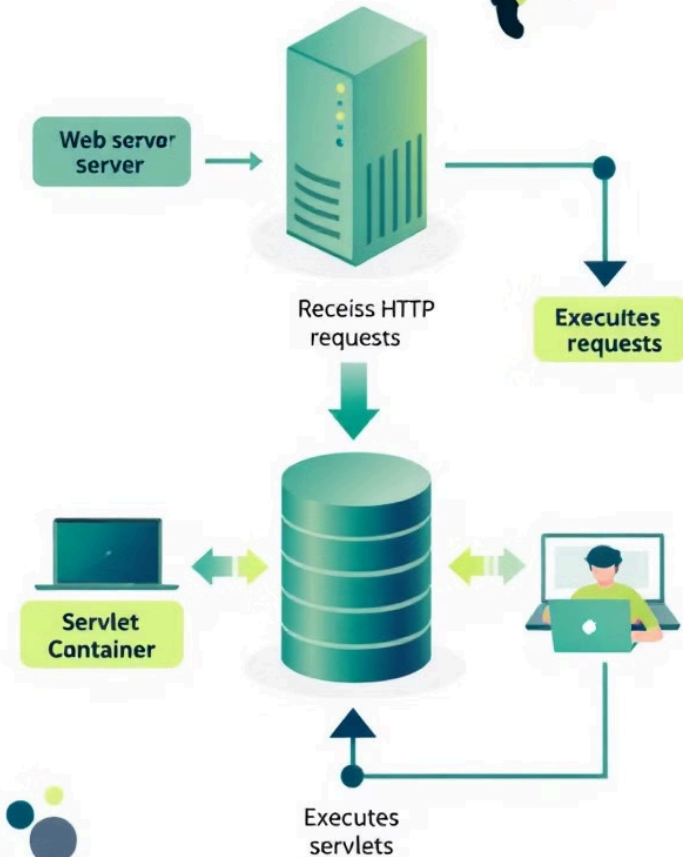


Mastering Java Servlets: Architecture, Lifecycle, and Web Applications

Servlets power enterprise Java applications by handling client requests on servers. They work within containers that manage their lifecycle and processing.

ISTM-WEB BASED PROGRAMMING

 by Taukir Alam



What is a Servlet?

Servlet Request-Response Process



Client Sends Request



Server Receives Request



Servlet Processes Request

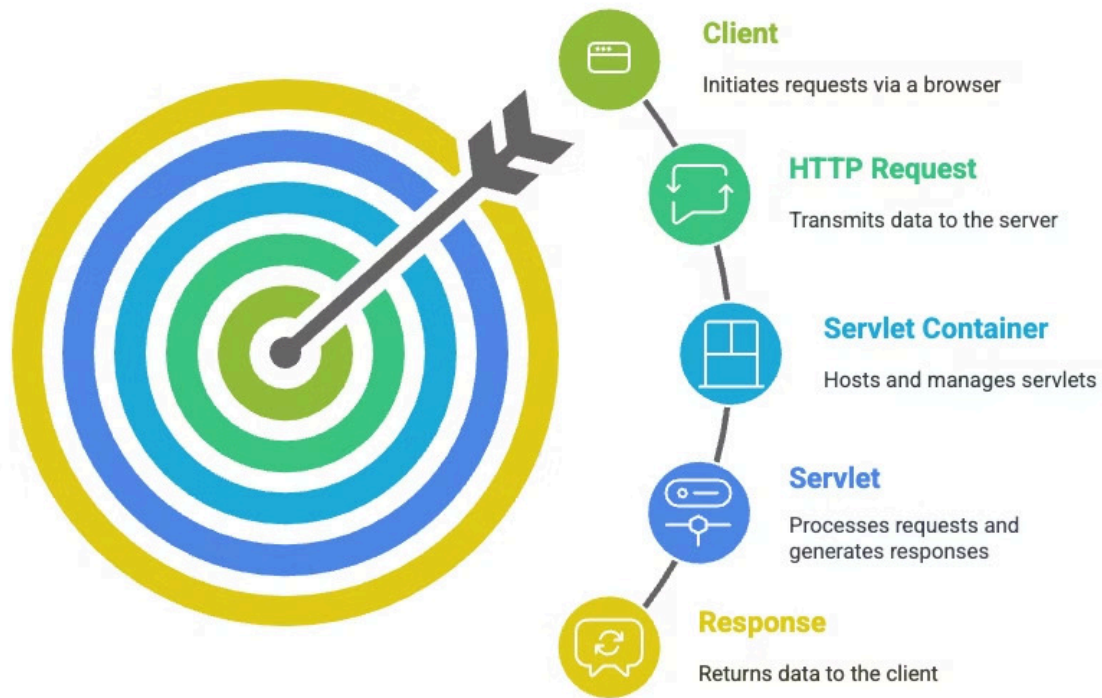


Servlet Generates Response



Server Sends Response

Java Servlet How it Works



Servlet Architecture



Client

Browser sends HTTP request



Web Server

Receives and forwards request



Web Container

Manages servlet lifecycle

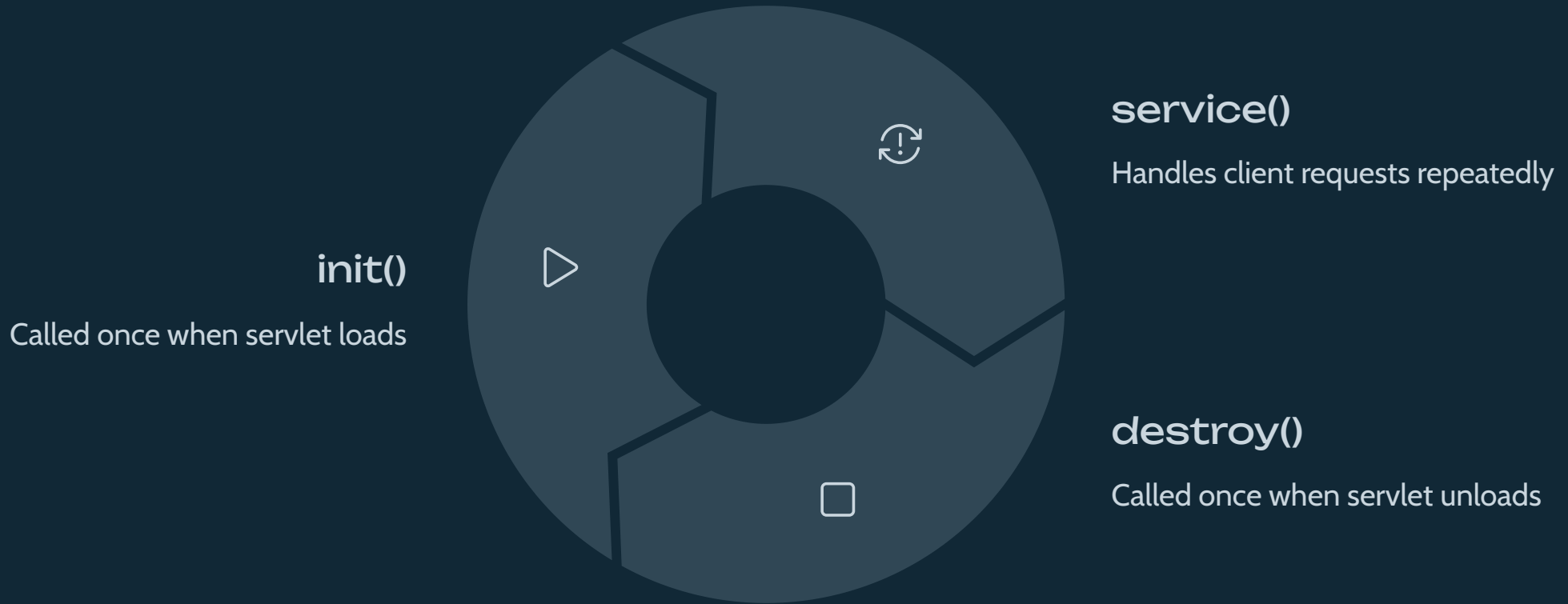


Servlet

Processes and returns response



Servlet Life Cycle



Form Handling: GET vs POST

GET Method

- Data appended to URL
- Limited to ~2048 characters
- Visible in browser history
- Bookmarkable

POST Method

- Data in request body
- No size limitation
- Not visible in URL
- More secure for sensitive data

Processing Form Data in Servlets

Retrieve Parameter

String username = request.getParameter("username");

Validate Input

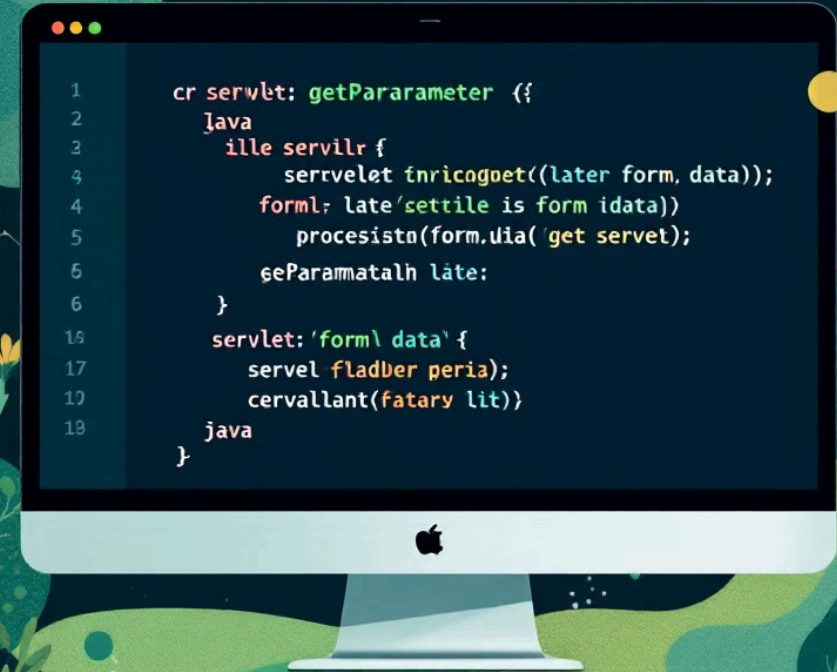
Check for null values and validate format

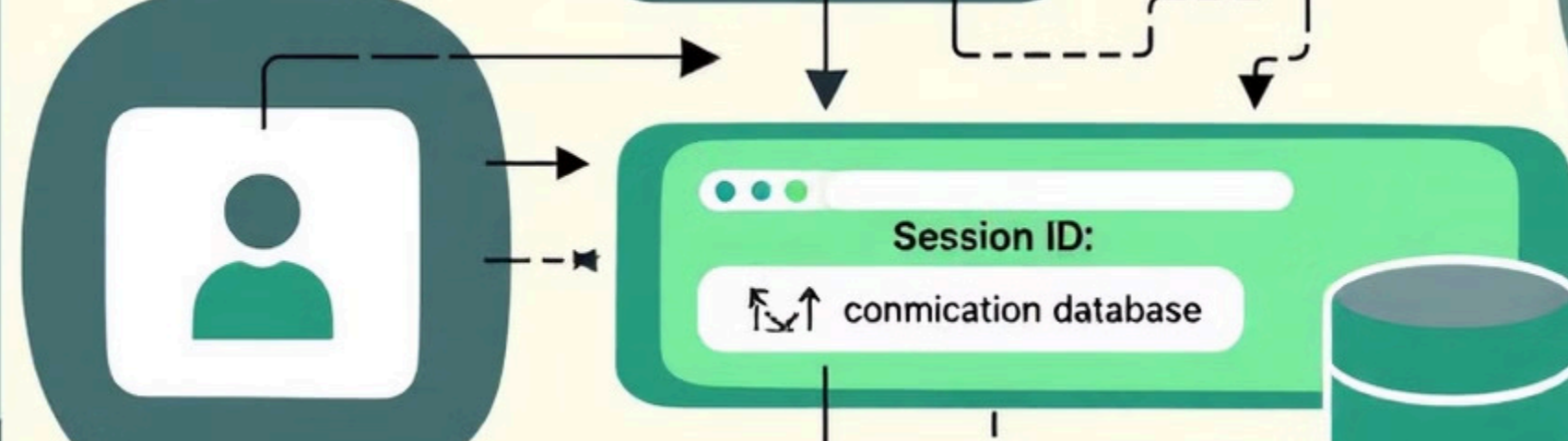
Process Data

Business logic, database operations

Generate Response

Create dynamic HTML with results





Session Handling in Servlets

Create/Get Session

```
HttpSession session = request.getSession();
```

Store Data

```
session.setAttribute("user", userObject);
```

Retrieve Data

```
User user = (User) session.getAttribute("user");
```

End Session

```
session.invalidate();
```

Understanding Cookies



Create Cookie

```
Cookie userCookie = new Cookie("username", "john");  
response.addCookie(userCookie);
```



Set Properties

```
userCookie.setMaxAge(3600); // 1 hour  
userCookie.setPath("/");
```



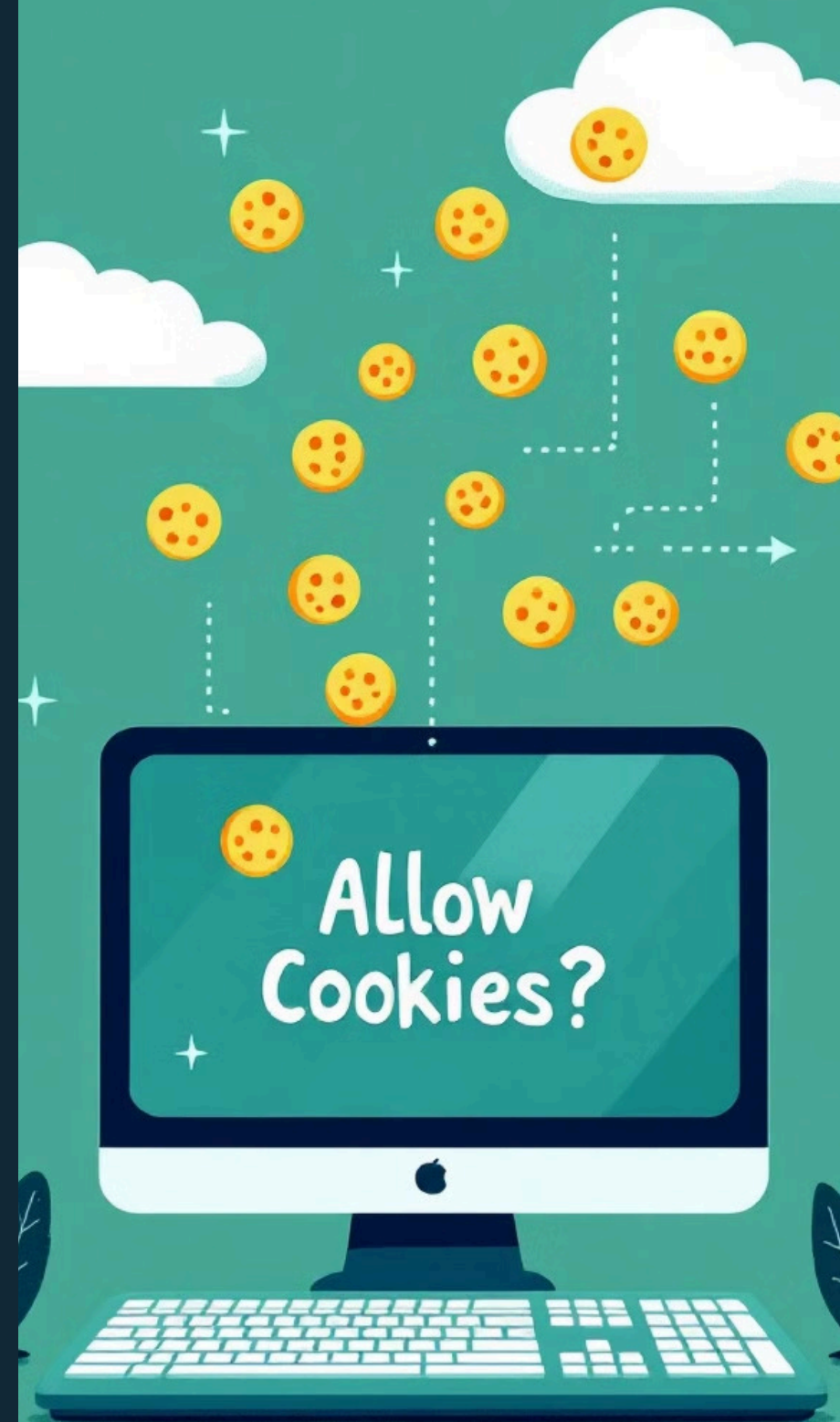
Read Cookie

```
Cookie[] cookies = request.getCookies();  
  
Loop through array to find by name
```

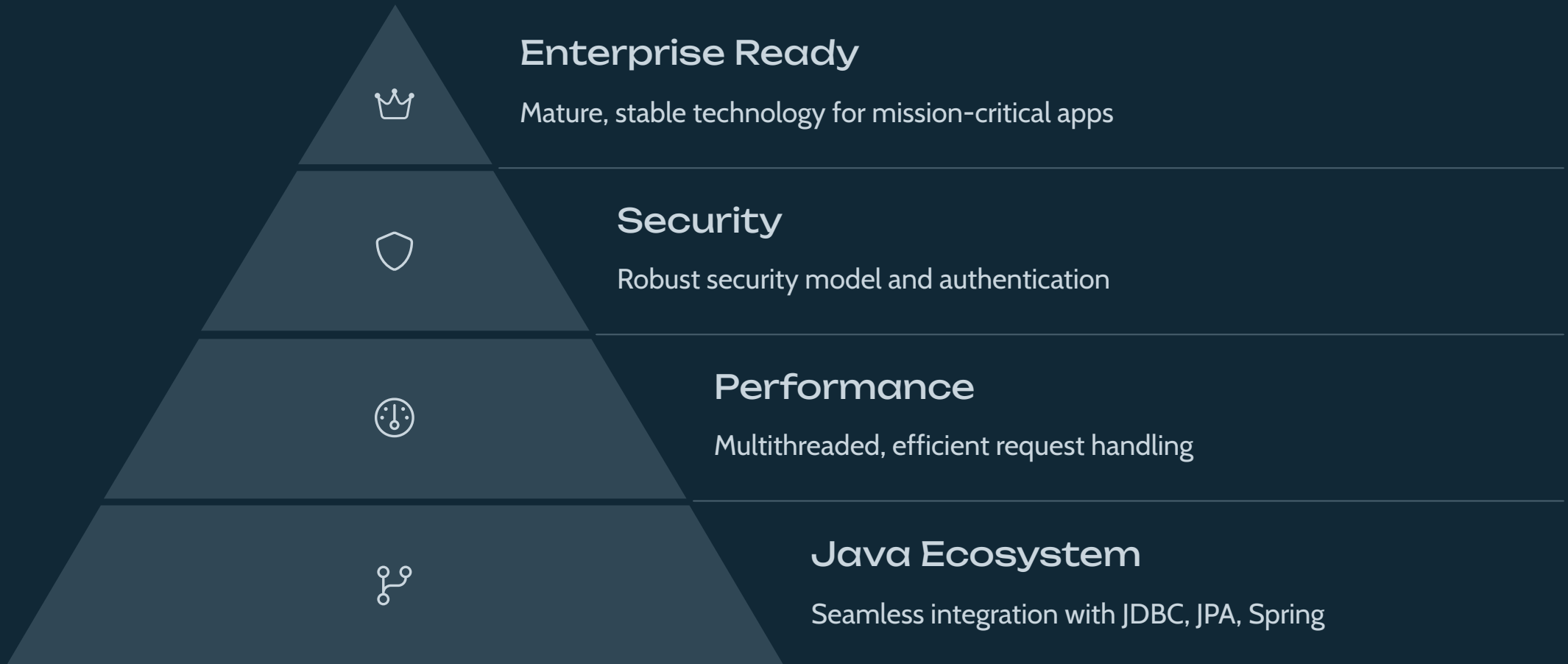


Delete Cookie

```
userCookie.setMaxAge(0);  
response.addCookie(userCookie);
```



Why Use Servlets?



Servlet Lifecycle in Web Applications



